# SOURAV PATTANAIK

National Institute of Science Education and Research (NISER), Jatni, India
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Life Science graduate with multidisciplinary research experience across systems biology, computational drug discovery, molecular biology, and chemistry. Experienced in developing computational workflows complemented by hands-on molecular biology and synthetic chemistry experience. Interested in integrated scientific approaches to understand complex biological systems and advance therapeutic discovery.

## EDUCATION

**2021 – 2026** **Integrated Master of Science in Biology | Minor degree in Chemistry**
National Institute of Science Education and Research (NISER), India
**Integrated CGPA: 8.40 / 10.0**
**Average Bachelor's (First Six semesters) SGPA: 8.02 / 10.0**
**Average Master's (Last four semesters) SGPA: 8.96 / 10.0**

**2021** **12th Boards (CBSE)** | DAV School — **94.6%**

**2019** **10th Boards (ICSE)** | Chinmaya Vidyalaya (E.M.) — **93.2%**

## PUBLICATIONS

**Peer-reviewed Publications**

Rana S., **Pattanaik S.** (co-first author), Thakur R., Joshi V., Jindal N., Tiwari R. R., & Sahoo G. C. (2026). Computational design and validation of a multi-epitope vaccine candidate against structural proteins of Langya virus. *Letters in Drug Design & Discovery*, 100296. https://doi.org/10.1016/j.lddd.2026.100296 **Published**

**Conference Abstracts**

**Pattanaik, S.**, Konkimalla, V. B. Directional Biological Network (DBN) Analysis Pro: Automated DBN construction, target prioritization, and pathway enrichment analysis. Presentation at *IndiBioCom 2026*. **Accepted**

## RESEARCH EXPERIENCE

**2024 – 2026** **MASTER'S THESIS** | Supervisor: Dr. V Badireenath Konkimalla, NISER, India
*"Integrating network-based drug discovery and stimulus-responsive drug carrier for treating testicular cancer through an in silico approach."* **Project Coursework Average across four semesters - 9.25/10**

- Developed DBN Analysis Pro, a web tool to integrate transcriptomic, pathway, and protein-protein interaction data for mechanistic prioritization of therapeutic targets.
- Designed a novel dual-stimulus-responsive drug carrier using Intrinsically Disordered Proteins(IDPs) and studied its conformational dynamics using CG simulations.
- Leveraged atomistic simulations for studying protein-ligand interactions after virtual screening for a potential drug molecule among 1.8 million molecules in the PubChem Database.
- Methods: AlphaFold, GROMACS, CG (MARTINI)/Atomistic (CHARMM) MD Simulation, Pharmacophore modeling, Protein-DNA Docking, Network analysis.

**May – July 2025** **SUMMER FELLOWSHIP** | Dr. Anbazhagan Kolandaswamy, RRMCH Bengaluru
*"Discovery of Biomarkers for Sepsis Disease through comparative analysis of gene expression profiles."*

- Optimized RNA extraction protocols and processed 52 patient blood samples, while maintaining high-integrity RNA extraction for quantitative profiling of 15 genetic targets(mRNA/miRNA).
- Identified 6 statistically significant potential biomarkers( P-values $< 0.05$).
- Methods: qPCR, RT-PCR, Biostatistical analysis, Flow cytometry for immunophenotyping, FlowJo software for analysis, gel electrophoresis.

**Jan – Mar 2025** **REMOTE COLLABORATION** | Dr. Karoline Faust, KU Leuven
*"Beta-testing of a new database associated with Systems Microbiology."*

- Extracted, structured, and interpreted growth data of 10 different bacterial species.
- Contributed to Beta-testing of the new database.
- Acknowledged in research pre-print (doi:10.1101/2025.10.13.682118).

**May – July 2024** **SUMMER INTERNSHIP** | Dr. Sindhuprava Rana, ICMR-NIREH Bhopal
*"Multi-Epitope Vaccine design against the structural proteins of Langya Virus."*

- Leveraged immunoinformatics to design a multi-epitope vaccine targeting 5 structural proteins of the Langya Virus.
- Suggested a novel strategy to include epitopes in the construct while maintaining solubility and biocompatibility and published the article as a co-first author.

**Nov – Dec 2023** **WINTER INTERNSHIP** | Dr. Abdur Rehman, Biochemistry lab, NISER
*"Introduction to Life Cycle of Tetrahymena."*

- Learned confocal microscopy and cell imaging using ImageJ.

**May – July 2023** **SUMMER SCHOOL** | Dr. S Mirza, IIT Indore
*"Principles of Drug Discovery and Development"*

- Focus: Understanding the activity of anti-inflammatory drugs and stages of drug discovery.

**May – July 2022** **SUMMER INTERNSHIP** | Dr. C S Purohit, Synthetic Chemistry Lab, NISER
*"Multi-step Organic synthesis of 2,6-Bis{[bis(2-pyridinylmethyl)amino]methyl}-4-(1,1-dimethylethyl)phenol by aminomethylation."*

- Methods: Multi-Step Synthesis, TLC, Column Chromatography, NMR.

## CONFERENCES AND CONCLAVE

- **Conferences:**
  Quarterly Union for Academic Disclosure (QUAD)
  IAS-SRFP 36$^{th}$ Mid-Year meeting 2025
  InBix 2025.
- Participated in **the Springer Nature India Research Conclave** (2023)

## RELEVANT TECHNICAL SKILLS

- **Computational Structural Biology:** mRNA vaccine design, Molecular docking, MD simulation (CHARMM/MARTINI), Molecular modeling, AI-based protein modeling, Pharmacophore modeling.
- **Bioinformatics:** Transcriptomics, Proteomics, Network biology, Multi-omics
- **Programming:** Python, LaTeX, Java, LINUX, GPU-based computing, Web tools development.
- **Wet-lab:** qPCR, RT-PCR, Gel electrophoresis, Plasmid/RNA/DNA isolation, Flow-cytometry, Basic X-ray crystallography, Protein purification, Synthesis, NMR, Column chromatography.

## AWARDS, FELLOWSHIPS AND SCHOLASTIC ACHIEVEMENTS

- **Competitive Exams Qualified:** National Entrance Screening Test 2021 [All India Rank (AIR) 263]; Graduate Aptitude Test in Engineering 2024 [AIR 1173]; Odisha Joint Entrance Examination 2021 [State Rank 3]; Odisha University of Agriculture and Technology 2021 [AIR 81]; National Eligibility cum Entrance Test 2021.
- **Scholarships/Fellowships:** Department of Atomic Energy (DAE)-DISHA Scholarship of 400000 INR(2021-2026); Indian Academy of Sciences-Summer Research Fellowship 30000 INR(May – July 2025).
- **Competitions:** Finalist of BENSPARK Ideathon, IISER Pune [2025]; Excelling ranks in Olympiads (English and Science) [2017-2019]; Won multiple inter-school and intra-college quizzes.

## LEADERSHIP AND RESPONSIBILITY

- **Gym Treasurer (2024-2025):** Handled expenditure and bills amounting to almost 50000 INR for NISER Gym Club; Conducted Power-lifting events and served as a coordinator for multiple events.
- **Student Representative:** Member of the Disciplinary Action Committee (DAC) for Students (2025-2026). Contributed actively to policy making and adjudicated cases like substance abuse and harassment to ensure a safe campus environment.

## HOBBIES AND EXTRA-CURRICULAR

- **Sports:** Represented NISER in Basketball (IISM 2023) and won inter-batch Basketball tournament (2024). Represented batch in Football (2024) and Badminton (2025) while winning bronze medals.
- **Cultural:** Won multiple dance and slogan writing competitions.
- **Other Interests:** Gym; Reading about novel therapeutic strategies in free time.

## REFEREES

**Dr. Sindhuprava Rana**
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